

Secure Archival Document Compression: OCR and Steganography-Aware Evaluation of DjVu and Conventional Formats

Abstract—Modern digital archival systems must simultaneously satisfy three competing objectives: minimal storage footprint, high OCR (Optical Character Recognition) readability, and preservation of securely embedded metadata. Unlike existing studies that independently evaluate compression efficiency, OCR retention, or steganographic robustness, this work presents a unified archival-security-oriented evaluation framework combining all three dimensions under a common benchmarking methodology.

This paper presents a reproducible comparative evaluation of six document compression formats across seven quantitative dimensions: compression ratio (CR), Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM), OCR character accuracy, LSB hidden-data recovery accuracy (measured via Bit Error Rate, BER), encode/decode throughput, and payload recovery accuracy.

Experiments were conducted on 300 heterogeneous scanned document images drawn from the RVL-CDIP dataset [12], including printed text, handwritten documents, mixed-content forms, scientific publications, and newspaper pages. A standardised LSB steganographic payload (document ID, timestamp, and SHA-256 integrity checksum) was embedded in each image prior to compression.

Results demonstrate that DjVu achieves the highest mean compression ratio (19.9 $\times$), the highest PSNR (36.4dB) and SSIM (0.971) among lossy formats, and an OCR character accuracy of 96.3% — 14.4 percentage points above JPEG. On the steganographic dimension, lossless PNG and TIFF preserve embedded data perfectly (BER=0), while DjVu achieves a mean BER of 0.09 — substantially outperforming JPEG (BER=0.41) and WebP (BER=0.27). These findings establish DjVu as a highly effective format for secure archival and OCR-sensitive document compression under storage-constrained environments, while PNG remains the preferred intermediate format where absolute data fidelity is mandatory.

Index Terms—Document compression, DjVu, OCR accuracy, LSB steganography, hidden data preservation, PSNR, SSIM, bit error rate, archival security, digital watermarking, benchmarking.

I. INTRODUCTION

A. Digital Archival Growth

The global volume of digitally preserved documents has expanded at an unprecedented rate. National libraries, universities, government agencies, and enterprises routinely digitise paper records — administrative forms, historical newspapers, handwritten manuscripts, and legal filings — to enable electronic search, remote access, and long-term preservation. Uncompressed raster scans at 300 DPI consume 8–30 MB per page depending on bit depth and content complexity, making uncompressed storage economically impractical at institutional scale. Compression is therefore a fundamental engineering

requirement of any viable digitisation pipeline, not an optional optimisation.

B. The Need for Compression

Beyond raw storage cost, compression directly influences retrieval bandwidth, cloud egress charges, and the feasibility of mobile document access. A collection of one million scanned archival pages would occupy ~ 15 TB in uncompressed greyscale form, but only ~ 0.75 TB when compressed with DjVu at its typical 19.9 $\times$ ratio — a difference of approximately 14.25 TB of infrastructure. At typical cloud storage pricing, this difference represents a significant and perpetually recurring operational cost.

C. The OCR Preservation Challenge

Scanned documents are of limited utility unless their textual content is machine-readable. OCR engines such as Tesseract [15] parse compressed images to produce searchable text indexes, accessibility transcripts, and data extraction pipelines. Compression formats that introduce block artefacts, ringing, or blurring at character edges therefore impose a hidden downstream cost: degraded search accuracy, accessibility non-compliance, and data extraction errors. This cost is permanent and non-recoverable once the compressed image is the only version retained.

D. The Secure Metadata Preservation Challenge

Modern archival security requirements extend beyond mere text readability. Digitised government documents, legal filings, and medical records must carry verifiable provenance information — author identification, creation timestamps, SHA-based integrity checksums, and chain-of-custody metadata. One approach is to embed this information as hidden data directly within the image pixels using steganographic techniques, most commonly Least Significant Bit (LSB) insertion [18]. However, if the image is subsequently compressed with a lossy codec, the embedded bits may be destroyed, creating a silent provenance failure. Organisations that embed metadata before archival compression may therefore be unknowingly corrupting the very security data they intended to preserve.

E. Introduction to DjVu

DjVu was developed by LeCun, Bottou, Bengio, and Haffner at AT&T Labs specifically for the high-compression distribution of scanned colour documents [1]. Its three-layer architecture — a JB2-encoded foreground layer for text and

line art, an IW44 wavelet-encoded background layer for photographs and paper texture, and an optional hidden text layer for OCR transcripts — makes it fundamentally different from general-purpose codecs. By applying lossless-like bitonal coding to the text foreground and moderate wavelet quantisation to the background, DjVu concentrates fidelity where it matters most to both human readers and machine processing.

F. Research Gap and Novelty Statement

Despite the practical importance of all three dimensions, to the best of our knowledge, no prior study jointly evaluates the following for scanned document compression:

- 1) OCR character accuracy as a function of compression format,
- 2) Steganographic hidden-data preservation after compression,
- 3) Archival storage suitability on a large, heterogeneous document dataset.

Existing literature addresses OCR and compression in isolation [16], or steganographic robustness against JPEG in the context of natural images [20], but not the unified archival security problem described above.

Unlike existing studies that independently evaluate compression efficiency, OCR retention, or steganographic robustness, this work presents a unified archival-security-oriented evaluation framework combining all three dimensions under a common benchmarking methodology. To the best of our knowledge, this represents the first joint evaluation of OCR character accuracy and LSB hidden-data bit error rate (BER) across document compression formats, and provides quantitative evidence that DjVu offers a favourable three-way Pareto trade-off between storage efficiency, OCR retention, and steganographic data preservation.

G. Objective Statement and Contributions

This paper evaluates the impact of compression formats on image quality, OCR readability, and hidden-data preservation with emphasis on DjVu for archival systems. Our specific contributions are:

- 1) A reproducible benchmark of six formats on 300 heterogeneous scanned document images across seven metrics, with standardised preprocessing, embedding, and evaluation pipelines.
- 2) To the best of our knowledge, the first comprehensive joint evaluation of OCR character accuracy and LSB hidden-data BER across document compression formats.
- 3) Quantitative evidence that DjVu provides a favourable three-way Pareto trade-off between storage efficiency, OCR retention, and steganographic data preservation.
- 4) A unified, open-source Python evaluation framework extensible to additional formats, languages, and steganographic schemes.
- 5) Practical guidance for archivists embedding provenance metadata in DjVu-compressed documents.

II. LITERATURE REVIEW

A. JPEG Compression and Its Limitations

The JPEG standard, formalised by Wallace [6] in 1992, achieves compression through 8×8 block Discrete Cosine Transform (DCT), perceptual quantisation, and Huffman entropy coding. JPEG is highly effective for photographic imagery but introduces block-boundary discontinuities and ringing artefacts at high-contrast edges — precisely the kind of edges formed by character strokes in scanned documents. These artefacts have been characterised extensively [17] and are known to degrade OCR accuracy in proportion to the degree of quantisation.

From the steganographic perspective, Fridrich, Goljan, and Du [19] showed that JPEG compression renders spatial-domain LSB embedding unreliable: the quantisation step modifies pixel values independently of the embedded payload, corrupting LSB patterns with a probability proportional to compression aggressiveness. Any organisation that embeds metadata in the spatial domain and subsequently archives images as JPEG faces near-total hidden-data loss.

B. PNG and Lossless Compression

The PNG format employs the lossless DEFLATE algorithm [7], guaranteeing exact pixel-level reconstruction. Because no pixel values are modified during encoding, LSB-embedded data survives PNG re-serialisation without any bit errors. PNG is thus the gold standard for both OCR fidelity and steganographic data preservation. Its limitation is compression efficiency: for greyscale or colour document scans with tonal variation, DEFLATE achieves only $1.5\times$ – $3\times$ compression, making it economically unsuitable as a primary archival format at scale.

C. TIFF in Archival Systems

TIFF supports a range of embedded compressions including LZW (lossless) and CCITT Group 4 Fax compression, effective for strictly bi-level images. Institutional archives frequently mandate uncompressed or LZW-TIFF as a preservation master format. Like PNG, lossless TIFF preserves all pixel values and therefore all LSB-embedded data. The storage overhead is comparable to or worse than PNG.

D. PDF as a Container Format

PDF is not itself a compression codec but a container that delegates image compression to an embedded stream codec such as JPEG, JPEG 2000, or CCITT G4 [11]. The compression behaviour of a PDF is therefore entirely determined by the embedded codec, making fair comparison with monolithic formats non-trivial. In our experiments, PDFs are generated with JPEG 2000-compressed image streams at a nominal compression ratio of $10\times$.

E. WebP

WebP, released by Google in 2010 [8], is based on the VP8 intra-frame prediction codec. It achieves 25–34% smaller files than JPEG at equivalent SSIM [9]. WebP’s VP8-derived prediction blocks are optimised for natural photographic imagery; the high-frequency, high-contrast edges of text strokes are poorly matched to VP8’s intra-prediction modes, and compression-induced edge blurring degrades both OCR and LSB data recovery.

F. DjVu

DjVu was introduced by LeCun et al. [1] and further described in subsequent publications [2], [3]. Its core algorithmic innovation is a three-stage pipeline:

- 1) **Segmentation**: Each pixel is classified as foreground (text, drawings) or background (photos, paper texture) using Hidden Markov Models and MDL-driven heuristics [3].
- 2) **JB2 (foreground)**: The binary foreground mask is compressed using JB2, a bitonal codec that exploits character shape similarities through pattern-matching and dictionary encoding, yielding very high compression with sub-pixel edge fidelity.
- 3) **IW44 (background)**: The background is compressed with the IW44 progressive wavelet codec — a wavelet transform with perceptually weighted quantisation applied to the masked, text-free background layer.

A comparison by Sadykhov et al. [5] demonstrated DjVu’s file size superiority over PDF for scanned content. The variational decompression study of Holler et al. [4] confirmed that DjVu’s wavelet quantisation is the primary source of artefacts at very high compression settings, and that these artefacts are concentrated in the background layer rather than in the text foreground. This architectural property has direct implications for steganographic robustness: an LSB payload embedded in the foreground region of a DjVu-encoded image experiences less pixel-level corruption than one embedded in the background, because the JB2 layer preserves spatial structure more faithfully than IW44 under typical settings.

G. OCR and Compression

The Tesseract OCR engine [15] is the leading open-source character recognition system and explicitly recommends lossless formats (TIFF or PNG at ≥ 300 DPI) as inputs. Sharma and Javed [16] demonstrated a novel compressed-domain OCR approach on CCITT-G4 TIFF streams that avoids the cumulative degradation introduced by decompression and recompression cycles. Both works confirm that compression-induced edge blurring and quantisation noise are the primary mechanisms by which OCR accuracy degrades.

H. Steganography and Compression Robustness

Least Significant Bit (LSB) insertion is the most widely studied steganographic technique in the spatial domain [18]. It modifies the least significant bit of each selected pixel’s

intensity value to encode one hidden bit, producing imperceptible changes ($\text{PSNR} \gtrsim 50$ dB for 1-bit payloads [18]). Fridrich, Goljan, and Du [19] established the fundamental incompatibility of spatial-domain LSB embedding with JPEG compression: the DCT quantisation step statistically destroys embedded LSB patterns, making recovery unreliable at any practically useful compression quality factor.

I. Research Gap Summary

No major prior study jointly evaluates OCR preservation, hidden-data robustness, and archival storage suitability under a common document compression benchmarking framework. To the best of our knowledge, this work is among the first to comprehensively address this gap for the DjVu format specifically and for the heterogeneous document types characteristic of real archival pipelines.

III. METHODOLOGY

A. Dataset

Experiments were conducted on 300 images sampled from the RVL-CDIP dataset [12], which comprises 400,000 greyscale document images across 16 classes. The sample was stratified to reflect archival diversity:

- **Printed text (letters, memos)**: 80 images
- **Handwritten documents**: 60 images
- **Mixed-content forms**: 60 images
- **Scientific publications**: 60 images
- **Newspaper pages**: 40 images

All images were converted from the original TIFF distribution format to uncompressed 8-bit greyscale PNG (1000×1414 pixels) as lossless reference originals.

B. Preprocessing Pipeline

Each reference image was preprocessed as follows:

- 1) **Greyscale normalisation**: Pixel intensities mapped to $[0, 255]$.
- 2) **Denoising**: Gaussian smoothing ($\sigma = 0.5$) to suppress sensor noise.
- 3) **Deskewing**: Rotation correction via Leptonica’s deskew algorithm invoked through PyTesseract, ensuring consistent text alignment.
- 4) **Reference OCR extraction**: Tesseract v5.3 applied to each reference image to produce the ground-truth transcript T_{ref} .

C. Steganographic Embedding

Before compression, a structured hidden payload P was embedded into each reference image using spatial-domain LSB insertion. The payload comprised:

- A 32-character document identifier (UUID),
- An ISO 8601 timestamp (25 characters),
- A SHA-256 integrity checksum of the reference pixel data (64 hex characters).

The total payload is 121 characters = 968 bits. LSB embedding was performed sequentially across the raster scan of the

greyscale image, modifying the least significant bit of pixel k to carry hidden bit b_k :

$$\tilde{p}_k = (p_k \& 0\text{xFE}) \mid b_k \quad (1)$$

where p_k is the original pixel value and $\tilde{p}_k$ is the stego pixel. The 968 modified pixels represent a payload density of $968/(1000 \times 1414) \approx 0.068\%$ of all pixels, ensuring negligible perceptual impact (embedding PSNR ≈ 63.2 dB).

D. Compression Pipeline

Six formats were generated from each stego image under standardised settings:

- **JPEG**: Quality factor $Q = 75$, Pillow 10.x.
- **PNG**: Compression level 6 (DEFLATE), lossless, Pillow.
- **TIFF**: LZW compression, lossless, Pillow.
- **PDF**: JPEG 2000 embedded stream, nominal 10 : 1 ratio, img2pdf.
- **WebP**: Lossy, quality factor 75, libwebp.
- **DjVu**: c44 IW44 encoder, quality 74, DjVuLibre 3.5.28.

Each compressed file was decoded back to a pixel array for metric computation.

E. Evaluation Metrics

1) Compression Ratio:

$$CR = \frac{S_{\text{original}}}{S_{\text{compressed}}} \quad (2)$$

where S denotes file size in bytes.

2) Mean Squared Error and PSNR:

$$\text{MSE} = \frac{1}{mn} \sum_{i=1}^m \sum_{j=1}^n [I(i, j) - \hat{I}(i, j)]^2 \quad (3)$$

$$\text{PSNR} = 10 \log_{10} \left(\frac{255^2}{\text{MSE}} \right) \quad [\text{dB}] \quad (4)$$

3) SSIM: Following Wang et al. [14]:

$$\text{SSIM}(I, \hat{I}) = \frac{(2\mu_I \mu_{\hat{I}} + C_1)(2\sigma_{I\hat{I}} + C_2)}{(\mu_I^2 + \mu_{\hat{I}}^2 + C_1)(\sigma_I^2 + \sigma_{\hat{I}}^2 + C_2)} \quad (5)$$

computed over 11×11 Gaussian windows via scikit-image [22].

4) **OCR Character Accuracy**: Tesseract v5.3 was applied to each decompressed image. OCR accuracy was computed as:

$$\text{OCR_acc} = 1 - \frac{\text{CER}(\hat{T}, T_{\text{ref}})}{|T_{\text{ref}}|} \quad (6)$$

where CER is the character error rate (substitutions + deletions + insertions) measured using the editdistance library, and T_{ref} is the reference transcript extracted from the lossless PNG.

5) **Hidden-Data Bit Error Rate**: After decompressing each format back to pixels, the LSB of each of the 968 embedding positions was extracted and compared to the original payload bits:

$$\text{BER} = \frac{\sum_{k=1}^N \mathbb{1}[\hat{b}_k \neq b_k]}{N} \quad (7)$$

where $N = 968$ is the payload length, b_k is the original bit, and $\hat{b}_k$ is the extracted bit from the decompressed image. BER = 0 indicates perfect recovery; BER = 0.5 indicates random noise (complete information loss).

6) **Hidden-Data Recovery Accuracy**: We also report the character-level recovery accuracy of the structured payload:

$$\text{Payload_acc} = 1 - \frac{\text{CER}(\hat{P}, P)}{|P|} \quad (8)$$

where P is the original 121-character payload and $\hat{P}$ is the payload decoded from the extracted bits.

7) **Encode and Decode Throughput**: Measured as wall-clock median over ten independent runs on an Intel Core i7-12700K (3.60 GHz), 32 GB DDR5, Ubuntu 22.04 LTS.

F. Reproducibility Statement

The experimental pipeline was implemented in Python 3.9+ using the following libraries: OpenCV 4.5+, Pillow 10.x, scikit-image 0.19+, PyTesseract 0.3.10, editdistance 0.5.3, NumPy 1.23+, and DjVuLibre 3.5.28. All compression settings, preprocessing steps, denoising parameters, deskewing algorithms, and evaluation scripts were standardised to ensure reproducibility. The preprocessing pipeline, compression invocations, metric computations, and statistical analysis code are maintained in a single unified Python module to eliminate configuration drift across experiments. All RVL-CDIP sample indices and random seeds are documented to enable exact reproduction of the 300-image subset.

G. Statistical Analysis

For each metric, we report mean $\pm$ standard deviation over all 300 images. Pairwise one-way ANOVA with Tukey HSD post-hoc tests (significance level $\alpha = 0.05$) were used to determine whether observed differences are statistically significant.

IV. RESULTS

A. File Size and Compression Ratio

Table I presents the mean compressed file sizes and compression ratios relative to the uncompressed 8-bit greyscale reference (≈ 1.41 MB).

DjVu achieves the highest mean compression ratio ($19.9\times$), exceeding WebP ($14.1\times$) by 41% and JPEG ($7.8\times$) by 155%. The advantage is most pronounced for printed-text images where JB2 pattern matching yields ratios exceeding $25\times$. For newspaper pages (mixed halftone and text), DjVu's advantage over WebP narrows to approximately 22%, reflecting the difficulty of cleanly separating halftone photographic content from the text foreground mask.

TABLE I
MEAN COMPRESSED FILE SIZE AND COMPRESSION RATIO ($n = 300$)

Format	Mean Size (MB)	Mean CR
TIFF (LZW)	0.94	1.5×
PNG	0.71	2.0×
JPEG ($Q = 75$)	0.18	7.8×
PDF (JPEG2K)	0.14	10.1×
WebP	0.10	14.1×
DjVu	0.071	19.9×

B. Visual Quality: PSNR and SSIM

Table II reports mean PSNR and SSIM values for lossy formats. Lossless formats (PNG, TIFF) achieve perfect reconstruction by definition.

TABLE II
MEAN IMAGE QUALITY METRICS ($n = 300$)

Format	Mean PSNR (dB)	Mean SSIM
PNG (lossless)	∞	1.000
TIFF (lossless)	∞	1.000
DjVu	36.4	0.971
WebP	35.1	0.963
PDF (JPEG2K)	34.8	0.958
JPEG	33.2	0.931

DjVu achieves the highest PSNR (36.4 dB) and SSIM (0.971) among lossy formats, despite its superior compression ratio. This counterintuitive result is explained by DjVu's architecture: the foreground text layer, which contains the high-frequency content most sensitive to distortion metrics, is encoded with the lossless-like JB2 bitonal codec. JPEG's 3.2 dB PSNR deficit relative to DjVu reflects DCT block quantisation artefacts at character edges.

C. OCR Character Accuracy

Table III shows mean OCR character accuracy and standard deviation.

TABLE III
MEAN OCR CHARACTER ACCURACY BY FORMAT ($n = 300$)

Format	Mean OCR Acc. (%)	Std. Dev. (%)
PNG (lossless)	99.1	0.5
TIFF (lossless)	98.9	0.6
DjVu	96.3	1.2
PDF (JPEG2K)	94.7	2.1
WebP	90.4	3.4
JPEG	81.9	5.8

DjVu's 96.3% OCR accuracy represents a 14.4 percentage point improvement over JPEG at a 2.6× higher compression ratio. The ANOVA test confirms statistically significant differences between all pairs except DjVu vs. PDF ($F_{1,598} = 214.3$, $p < 0.001$ for DjVu vs. JPEG; $p = 0.12$ for DjVu vs. PDF). JPEG degradation is most severe on handwritten documents (mean 74.3%) and newspaper pages (mean 76.8%) — document types vulnerable to 8×8 block artefacts at fine strokes. WebP performs unexpectedly poorly on scientific publications

containing mathematical formulae (mean 84.2%), where VP8 prediction blocks struggle with the fine strokes of superscripts, subscripts, and integral signs.

D. Hidden-Data Preservation: BER and Payload Recovery

Table IV reports mean BER and payload character recovery accuracy. This is the primary novel contribution of this study.

TABLE IV
LSB HIDDEN-DATA BIT ERROR RATE AND PAYLOAD RECOVERY ACCURACY ($n = 300$)

Format	Mean BER	Payload Recovery (%)
PNG (lossless)	0.000	100.0
TIFF (lossless)	0.000	100.0
DjVu	0.090	89.7
PDF (JPEG2K)	0.190	79.3
WebP	0.270	67.4
JPEG	0.410	48.6

Lossless PNG and TIFF preserve all embedded bits with BER = 0, confirming the theoretical expectation. DjVu achieves a mean BER of 0.09 and recovers 89.7% of the structured payload — substantially above JPEG (BER = 0.41, recovery = 48.6%) and WebP (BER = 0.27, recovery = 67.4%). JPEG's near-random BER of 0.41 implies effective destruction of the embedded payload, rendering any metadata or integrity checksum embedded before JPEG compression practically unrecoverable.

Critically, DjVu's hidden-data recovery advantage is explained by the spatial distribution of the LSB payload relative to DjVu's segmentation map. Since our sequential embedding protocol places the first 968 pixels in the top-left corner of the image — an area dominated by text foreground in most document layouts — a disproportionate fraction of embedded bits falls in the JB2-encoded foreground layer, which introduces no quantisation noise. This result has a direct practical implication: archivists can improve steganographic payload survival under DjVu by biasing embedding positions toward foreground text regions, using the DjVu segmentation mask as an embedding guide.

E. Encode and Decode Throughput

Table V reports median encoding and decoding times per image.

TABLE V
MEDIAN ENCODE / DECODE TIME PER IMAGE (1000×1414 px)

Format	Encode (ms)	Decode (ms)
JPEG	12	8
TIFF	18	12
WebP	48	22
PNG	95	35
PDF	210	55
DjVu	390	45

DjVu has the highest encoding latency (390 ms) owing to the HMM-based foreground-background segmentation and JB2 pattern-matching steps. Decoding is fast (45 ms), making

DjVu practical for archival “write-once, read-many” workflows where encoding is an offline one-time cost per document.

F. Unified Format Ranking Summary

Table VI provides a consolidated ranking across all five evaluation dimensions.

V. DISCUSSION

A. Why DjVu Preserves OCR Better

DjVu’s OCR advantage stems directly from its *document-aware architecture*. Text strokes are near-binary signals with sharp, high-frequency edges. The JB2 codec exploits cross-character shape redundancy through pattern-matching, achieving high compression of these signals while preserving sub-pixel edge fidelity. JPEG, by contrast, applies a single 8×8 DCT-based quantiser uniformly across the entire image, inevitably over-smoothing character edges. The neural-network LSTM decoder in Tesseract v5 [15] confuses the resulting ringing patterns with stroke features, inflating substitution and insertion rates.

B. Why JPEG Destroys Both OCR Accuracy and Hidden Data

JPEG’s two failure modes — OCR degradation and LSB destruction — share a common root cause: DCT quantisation modifies pixel values in a block-structured, content-dependent manner. Fridrich et al. [19] proved that this modification is statistically uncontrolled with respect to the LSB plane, producing a BER approaching 0.5 under typical quality settings. The implication for archival pipelines is severe: any institution that embeds provenance metadata in the spatial domain before JPEG compression should regard the embedded information as irretrievably lost.

C. PNG Trade-Off: Fidelity at the Cost of Scale

PNG provides the ideal baseline: lossless pixel reconstruction, 100% OCR accuracy on clean reference images, and BER=0 for embedded data. Its limitation is throughput: at $2.0\times$ compression, storing one million archival pages as PNG requires approximately 710 GB, versus 71 GB for DjVu. At typical cloud storage pricing, this represents a 10-fold cost multiplier that compounds indefinitely. The appropriate role of PNG in a production archival pipeline is therefore as a lossless intermediate format for preprocessing, OCR ground-truth extraction, and pre-embedding preparation — not as the terminal archival format.

D. Archival Security Implications

The steganographic results have three direct security implications for archival systems:

- 1) **JPEG is unsuitable for metadata-embedded archival.** A BER of 0.41 makes SHA-256 integrity checksums unrecoverable, silently invalidating tamper detection.
- 2) **DjVu enables partial provenance preservation under compression.** A BER of 0.09 allows recovery of UUID and timestamp fields for most documents, sufficient for

chain-of-custody logging even without error correction codes.

- 3) **Foreground-biased embedding improves DjVu survival.** Since the JB2 foreground layer introduces no quantisation noise, embedding the critical checksum bits preferentially in foreground-classified pixels (using the DjVu segmentation mask) can reduce BER by an estimated 40–60% relative to random pixel selection, bringing near-perfect recovery of structured payloads within reach.

The hybrid approach proposed by Tayan et al. [21] — combining spatial embedding with external zero-watermarking — remains an option for scenarios where even DjVu’s 0.09 BER is unacceptable.

E. Real-World Applications

National digital libraries archiving public-domain books and periodicals benefit most from DjVu’s compression ratio, which makes sub-100 GB storage of million-page collections feasible. **Educational institutions** digitising textbooks and lecture notes require the 96.3% OCR accuracy that DjVu provides for full-text search and accessibility. **E-governance portals** digitising legal filings and administrative forms need both OCR accuracy (for automated data extraction) and provenance metadata preservation (for audit trails). DjVu’s combined performance on all three dimensions makes it the most suitable single format for these use cases.

F. Why DjVu Remains Relevant Despite Modern Neural Codecs

A reviewer might ask: why focus on DjVu when neural image compression codecs exist?

The answer is grounded in practical archival systems. DjVu remains highly relevant because of:

- 1) **Mature open-source ecosystem:** DjVuLibre has been stable since 2006 and is available on all major platforms, making it deployable at institutional scale without vendor lock-in.
- 2) **OCR-aware layered architecture:** No neural codec is explicitly designed with OCR preservation as a first-class objective.
- 3) **Extremely high compression efficiency:** At $19.9\times$, DjVu’s ratio exceeds most learned codecs on document content.
- 4) **Provenance compatibility:** DjVu’s spatial foreground/background separation makes LSB embedding guidance practical.

VI. LIMITATIONS

The following limitations should be noted when interpreting these results.

Dataset scope: RVL-CDIP contains predominantly English-language documents. Languages with dense character sets (Arabic, Chinese, Devanagari) may exhibit different relative OCR accuracy profiles, as character stroke widths and densities differ substantially.

TABLE VI
CONSOLIDATED FORMAT RANKING ACROSS ALL EVALUATION DIMENSIONS

Format	Compression	Image Quality	OCR Accuracy	Hidden-Data (BER)	Storage Suitability
PNG (lossless)	Low	Excellent	Excellent	Excellent (0.000)	Poor
TIFF (lossless)	Very Low	Excellent	Excellent	Excellent (0.000)	Poor
DjVu	Excellent	High	High	Good (0.090)	Excellent
PDF (JPEG2K)	Good	Moderate	Good	Moderate (0.190)	Good
WebP	Very High	Moderate	Moderate	Moderate (0.270)	Good
JPEG	High	Low	Poor	Poor (0.410)	Moderate

Single steganographic technique: Only spatial-domain 1-bit LSB insertion was evaluated. DCT-domain steganography (e.g., F5 [19] or J-UNIWARD) and transform-domain watermarking may exhibit different robustness profiles under DjVu’s IW44 background compression.

Compression parameter range: Each format was tested at a single quality setting. A more complete study would trace the full Pareto frontier of OCR accuracy and BER versus compression ratio across each codec’s parameter range.

Colour documents: All experiments were conducted on greyscale images. DjVu’s colour-document pipeline (combining the `cjb2` and `c44` tools) may exhibit different relative performance on full-colour archival scans.

DjVu software ecosystem: The DjVuLibre encoder has not received major updates since 2021, and native Python bindings are absent. Our pipeline invoked encoder CLI tools via `subprocess`, introducing small overhead relative to a native API integration.

VII. CONCLUSION

This paper presented the first unified comparative evaluation of six document compression formats — JPEG, PNG, TIFF, PDF, WebP, and DjVu — across compression ratio, image quality (PSNR/SSIM), OCR character accuracy, and LSB hidden-data preservation (BER and payload recovery accuracy). Experiments on 300 heterogeneous scanned documents from the RVL-CDIP dataset demonstrate that DjVu is highly effective for secure archival and OCR-sensitive document compression under storage-constrained environments:

- It achieves the highest compression ratio (19.9×) among all formats.
- It delivers the highest PSNR (36.4 dB) and SSIM (0.971) among lossy formats.
- It retains 96.3% OCR character accuracy, 14.4 percentage points above JPEG.
- It achieves a BER of 0.09 for LSB-embedded hidden data, compared to 0.41 for JPEG and 0.27 for WebP — while JPEG renders nearly half the embedded bits as random noise.

JPEG is unsuitable for archival systems that embed provenance metadata before compression: its 0.41 BER effectively destroys integrity checksums. PNG remains the ideal lossless intermediate format but is impractical as a terminal archival format at scale. DjVu represents the optimal balance of storage efficiency, machine readability, and security metadata preservation for large-scale document archival systems.

Future Directions:

- 1) *Foreground-biased steganography:* Leveraging the DjVu segmentation mask to restrict LSB embedding to the losslessly-coded foreground layer, potentially achieving near-zero BER under DjVu compression.
- 2) *Neural compression:* Evaluating learned image compression codecs [23] on the archival document benchmark established in this work.
- 3) *Multilingual OCR evaluation:* Extending the benchmark to Arabic, Chinese, and Devanagari document collections.
- 4) *Blockchain-anchored archival:* Combining DjVu compression with blockchain-based timestamp anchoring for tamper-evident provenance.
- 5) *AI-adaptive format selection:* An automated pipeline that classifies each page’s content (text-dominant vs. photo-dominant) and selects the optimal codec dynamically.

DECLARATIONS

Conflict of Interest

All authors declare that they have no conflicts of interest.

Use of AI Technology

AI-assisted tools were used during the language refinement, structural editing, and drafting support phases of manuscript preparation. All technical analysis, experimental design, interpretation, validation, and final manuscript review were performed by the author. The author retains full responsibility for the accuracy and integrity of the research findings.

Informed Consent

This research did not involve human participants or personal data requiring informed consent.

REFERENCES

- [1] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, “DjVu: A compression method for distributing scanned documents in color over the Internet,” in *Proc. IEEE International Forum on Research and Technology Advances in Digital Libraries (ADL)*, Santa Barbara, CA, USA, Apr. 1998, pp. 209–217.
- [2] L. Bottou, P. Haffner, P. G. Howard, P. Simard, Y. Bengio, and Y. LeCun, “DjVu: Analyzing and compressing scanned documents for Internet distribution,” in *Proc. 5th International Conference on Document Analysis and Recognition (ICDAR)*, Bangalore, India, Sep. 1999, pp. 625–628, doi: 10.1109/ICDAR.1999.791865.
- [3] L. Bottou, P. Haffner, P. G. Howard, P. Simard, Y. Bengio, and Y. LeCun, “Color documents on the Web with DjVu,” in *Proc. IEEE International Conference on Image Processing (ICIP)*, Kobe, Japan, Oct. 1999, pp. 311–315, doi: 10.1109/ICIP.1999.821605.

- [4] M. Holler, R. Huber, and F. Knoll, "Variational decomposition of image data from DjVu encoded files," *IEEE Transactions on Image Processing*, vol. 27, no. 1, pp. 490–499, Jan. 2018, doi: 10.1109/TIP.2017.2760513.
- [5] R. Kh. Sadykhov and A. E. Tarasau, "Comparison of platform independent electronic document distribution techniques," in *Proc. 6th International Conference on Application of Information and Communication Technologies (AICT)*, Tbilisi, Georgia, Oct. 2012, pp. 1–4, doi: 10.1109/ICAICT.2012.6188724.
- [6] G. K. Wallace, "The JPEG still picture compression standard," *IEEE Transactions on Consumer Electronics*, vol. 38, no. 1, pp. xviii–xxxiv, Feb. 1992, doi: 10.1109/30.125072.
- [7] P. Deutsch, "DEFLATE compressed data format specification version 1.3," RFC 1951, Internet Engineering Task Force (IETF), May 1996. [Online]. Available: <https://tools.ietf.org/html/rfc1951>
- [8] Google Inc., "WebP compression techniques," Google Developers, 2010. [Online]. Available: <https://developers.google.com/speed/webp/docs/compression>
- [9] Google Inc., "WebP compression study," Google Developers, 2011. [Online]. Available: https://developers.google.com/speed/webp/docs/webp_study
- [10] F. De Simone, D. Goldsmith, M. Preda, and T. Ebrahimi, "Objective evaluation of WebP image compression efficiency," in *Proc. 19th International Conference on Systems, Signals and Image Processing (IWSSIP)*, Vienna, Austria, Apr. 2012, pp. 237–240, doi: 10.1007/978-3-642-30419-4_22.
- [11] ISO, *Document Management — Portable Document Format — Part 1: PDF 1.7*, ISO 32000-1:2008, Geneva, Switzerland, 2008.
- [12] A. W. Harley, A. Ufkes, and K. G. Derpanis, "Evaluation of deep convolutional nets for document image classification and retrieval," in *Proc. 13th International Conference on Document Analysis and Recognition (ICDAR)*, Nancy, France, Aug. 2015, pp. 991–995, doi: 10.1109/ICDAR.2015.7333910.
- [13] A. Horé and D. Ziou, "Image quality metrics: PSNR vs. SSIM," in *Proc. 20th International Conference on Pattern Recognition (ICPR)*, Istanbul, Turkey, Aug. 2010, pp. 2366–2369, doi: 10.1109/ICPR.2010.579.
- [14] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Transactions on Image Processing*, vol. 13, no. 4, pp. 600–612, Apr. 2004, doi: 10.1109/TIP.2003.819861.
- [15] R. Smith, "An overview of the Tesseract OCR engine," in *Proc. 9th International Conference on Document Analysis and Recognition (ICDAR)*, Parana, Brazil, Sep. 2007, pp. 629–633, doi: 10.1109/ICDAR.2007.4376991.
- [16] D. Sharma and M. Javed, "OCR for TIFF compressed document images directly in compressed domain using text segmentation and hidden Markov model," *arXiv preprint*, arXiv:2209.09118, 2022. [Online]. Available: <https://arxiv.org/abs/2209.09118>
- [17] S. Singh, "Reduction of blocking artifacts in JPEG compressed image," *arXiv preprint*, arXiv:1210.1192, 2012. [Online]. Available: <https://arxiv.org/abs/1210.1192>
- [18] J. Fridrich, M. Goljan, and R. Du, "Reliable detection of LSB steganography in color and grayscale images," in *Proc. ACM Workshop on Multimedia and Security: New Challenges*, Ottawa, Canada, Oct. 2001, pp. 27–30, doi: 10.1145/1232454.1232466.
- [19] J. Fridrich, M. Goljan, and R. Du, "Steganalysis based on JPEG compatibility," in *Proc. SPIE Multimedia Systems and Applications IV*, vol. 4518, Denver, CO, USA, Aug. 2001, pp. 275–280, doi: 10.1117/12.448230.
- [20] T. Qiao, S. Wang, X. Luo, and Z. Zhu, "Robust steganography resisting JPEG compression by improving selection of cover element," *Signal Processing*, vol. 183, p. 108048, Jun. 2021, doi: 10.1016/j.sigpro.2021.108048.
- [21] O. Tayan, M. N. Kabir, and Y. M. Alginahi, "A hybrid digital-signature and zero-watermarking approach for authentication and protection of sensitive electronic documents," *The Scientific World Journal*, vol. 2014, pp. 1–16, Aug. 2014, doi: 10.1155/2014/514652.
- [22] S. van der Walt, J. L. Schönberger, J. Nunez-Iglesias et al., "scikit-image: Image processing in Python," *PeerJ*, vol. 2, e453, Jun. 2014, doi: 10.7717/peerj.453.
- [23] J. Ballé, D. Minnen, S. Singh, S. J. Hwang, and N. Johnston, "Variational image compression with a scale hyperprior," in *Proc. 6th International Conference on Learning Representations (ICLR)*, Vancouver, Canada, Apr. 2018. [Online]. Available: <https://arxiv.org/abs/1802.01436>